

Juan David Munoz Bolanos

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at

March 19, 2026

ASML

Hiring Manager

De Run 6501

5504 Veldhoven

Netherlands

Subject: Application for Optical System Engineer

Dear Hiring Manager,

As a physicist and optical engineer with a strong background in high precision metrology, optical simulation, and system integration, I am highly motivated to apply for an Optical Engineering position at ASML. I am drawn to ASML's commitment to pushing the physical limits of lithography and precision engineering, and I am eager to apply my expertise in developing next-generation opto-mechatronic modules.

During my Ph.D. at the Medical University of Innsbruck, I developed a system-level understanding of complex precision optics. I designed and built an automated wavefront sensing and phase retrieval system from scratch, establishing sub-wavelength high precision metrology protocols. This involved the complete integration of hardware and software to carry the end of a successful project, ensuring every optical component meets strict performance criteria through rigorous ZEMAX OpticStudio tolerance analysis.

Before my Ph.D., I gained industrial machine vision experience at INTECOL S.A.S., deploying high-speed automated inspection systems, and R&D experience at IPHT Jena, building specialized Raman spectrometers. This combination of industrial vision integration, academic rigor in optical metrology, and practical engineering allows me to bridge the gap between design concepts and robust, automated optical instruments.

Having participated in the ASML Summer School, I am familiar with the complex and collaborative environment required to build advanced lithography systems. I am completing my Ph.D. this May and would welcome the opportunity to discuss how my hands-on optical engineering and metrology experience can contribute to your team.

Sincerely,

Juan David Munoz Bolanos

Optical Systems Engineer

Medical University of Innsbruck

Juan David Munoz Bolanos

Optical Systems Engineer

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at | LinkedIn Profile



PROFESSIONAL SUMMARY

Optical Systems Engineer specializing in high precision metrology, optical modeling, and opto-mechatronics. Delivering successful peer-reviewed projects, including integration of adaptive optics for diffraction-limited multi-photon microscopy and full-cycle development of 1064 nm fiber probe Raman spectrometers. ASML Summer School participant driven to contribute to state-of-the-art optical systems and yield alignment technologies.

TECHNICAL COMPETENCIES

High Precision Metrology & Simulation

Deep knowledge of imaging physics, custom Python imaging simulations (Zernike), and optical modeling. Familiar with Zemax OpticStudio for tolerancing custom dispersive probes and resolving complex phase aberrations.

Opto-Mechatronics & Automation

Proven ability to automate optical benchtop experiments and integrate hardware with software, including synchronizing fluidic tuneable lenses via custom Python, C++, and LabVIEW interfaces.

Machine Vision & Image Processing

Industrial deployment of machine vision pipelines (Halcon, OpenCV) for high-speed automated inspection, yield assessment, and defect detection.

System Integration & Architecture

End-to-end R&D experience bringing complex optical setups from concept to fully functional, calibrated prototypes in both academic and industrial settings.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

PhD Researcher - Optical Metrology & Precision Measurements

Innsbruck, Austria

Jan 2022 – Present

- **Adaptive Optics Integration (Biomedical Optics Express '23 / ACS Photonics '25):** Designed an automated wavefront sensing system using a custom C++/Python/LabVIEW control stack. Synchronized 63-actuator Deformable Phase Plates (DPP) and Texas Instruments Phase Light Modulators (TI-PLM) to correct aberrations, achieving diffraction-limited resolution in deep-tissue microscopy.
- **Interferometric Phase Retrieval & Modeling:** Engineered system-level wavefront sensing frameworks using F-SHARP interferometry and batched-DASH optimization. Performed optical simulations alongside Python Zernike decomposition, experimentally resolving spatial shifts via a novel "3N algorithm."
- **Project Management:** Managed the specification and procurement of high-precision opto-mechanical components, interferometers, and DAQ electronics, steering complex optofluidic setups to peer-reviewed completion.

Leibniz Institute of Photonic Technology (IPHT)

R&D Researcher (Photonics & Data)

Jena, Germany

Jun 2020 – Dec 2021

- **Custom Spectrometer Prototyping (Applied Sciences '24):** Led the optical design, tolerancing, and construction of a dispersive 1064 nm fiber probe Raman spectrometer for clinical applications, handling rigorous detector alignment and calibration.
- **Data Analysis & AI:** Applied supervised machine learning to 2D hyperspectral Raman images from the custom probe, achieving high-accuracy tissue classification and accelerating characterizations.

INTECOL S.A.S.

Medellín, Colombia

- **Industrial Vision Deployment:** Deployed Halcon-based image processing pipelines for high-speed manufacturing inspection, resolving production failures through custom illumination and optics.
- **System Architecture:** Selected cameras, sensors, and illumination schemes to fully implement and validate optical inspection setups on active production lines.

EDUCATION

Medical University of Innsbruck

Austria

Ph.D. in Optical Systems Engineering (Adaptive Optics & Microscopy)

Jan 2022 – Present

Friedrich Schiller University Jena

Germany

M.Sc. in Photonics

2019 – 2021

University of Cauca

Colombia

B.Sc. in Physics Engineering

2012 – 2018

PUBLICATIONS & AWARDS

- **Awards and Selections:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026 for best Colombian professionals.
- 'Juan David Munoz Bolanos et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).
textitWork Done': 'Integrated adaptive optics to achieve diffraction-limited resolution in deep-tissue imaging, resolving phase aberrations experimentally.'
- 'Juan David Munoz Bolanos et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. Applied Sciences, 14(11), 4726 (2024).
textitWork Done': 'Concepted and developed a custom dispersive spectrometer probe, covering full optical design, tolerancing, and detector calibration.'
- 'Sohmen M., Juan David Munoz Bolanos et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).
textitWork Done': 'Integrated tuneable lenses into a scanning microscope and built the software interface for fast, aberration-free focusing.'

LANGUAGES

English (Fluent / Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte

Medical University of Innsbruck

Head of Institute for Biomedical Physics

monika.ritsch-marte@i-med.ac.at

Dr. Christoph Krafft

Leibniz Institute of Photonic Technology

Group leader for Spectroscopy Imaging

christoph.krafft@leibniz-ipht.de

AGREEMENT FOR GUESTS

to carry out a practical training
with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020


Prof. Dr. Popp
Chairman
of Leibniz-IPHT


F. Sonderrmann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

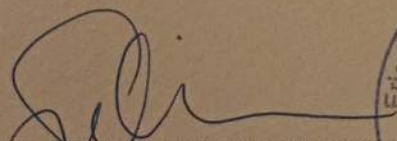
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
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OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garcés
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stämmler

Dr. Thomas Stämmler
CTO & Head of Product Strategy



Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Passports № / Passport No

MUNOZ BOLANOS

COLOMBIAN

Sexo / Sex
Lugar de nacimiento / Place of birth

11 DIC/DEC 2018

Fecha de vencimiento / Date of expiry

10 DIC/DEC 2028

G. ANTIOQUIA

Forma del bulbo / *bulbosus* & *bulbosus*

Dear Family

[illegible]

